TECHNICAL SPECIFICATION & HARDWARE BLUEPRINT

Agentic RAG Researcher Model Architecture

System Class:	Dense Transformer + Tool-Calling Agent	**Version:**	v3.2-Enterprise
Context Window:	128k Tokens (RoPE Embeddings)	**Latency Target:**	< 450ms First Token Time (TTFT)

1. Overview & Architectural Stack

The **Agentic RAG Researcher** architecture combines a dense decoder-only Transformer foundation model with an autonomous planning loop and dual-stage dense/sparse retrieval mechanisms. The system autonomously selects retrieval tools, validates structured schemas (e.g., CSV, SQL, PDF parsing), and performs iterative context expansion before generating final cited outputs.

2. Core Model Parameters & Hyperparameters

Hyperparameter / Layer Component	Specification Value	Notes / Configuration
Total Parameters	8.2 Billion	FP16 / INT4 Quantized Deployment
Attention Mechanism	Grouped-Query Attention (GQA)	8 Key-Value Heads, 32 Query Heads
Positional Embeddings	Rotary Position Embedding (RoPE)	Theta = 500,000 (Supports 128k Context)
Activation Function	SwiGLU	Hidden dimension scaled to 14,336
Embedding Model (Vector Search)	text-embedding-3-large / BGE-M3	1,536-dimensional Dense Vector Index

3. Agentic RAG Control Loop

The model executes a 4-stage internal execution pipeline to eliminate hallucination and resolve missing schema fields during query time:

1. **Query Decomposition & Intent Routing:** Parses incoming complex prompts into structured API tool calls or vector similarity searches.
2. **Retrieved Context Verification:** Evaluates retrieved structured/unstructured snippets against target schemas (e.g., verifying `churn_risk_score` presence).
3. **Correction & Schema Self-Healing:** If critical columns or metadata are absent, triggers fallback search modules or alerts for complete CSV ingestion.
4. **Synthesized Response Generation:** Formulates deterministic, grounded answers complete with inline source citations and numerical tables.

4. Vector Store & Dense Indexing Pipeline

```
[User Query]
     │
     ▼
┌────────────────────────────┐
│ Router & Intent Parser     │
└────────────┬───────────────┘
             │
      ┌──────┴──────────────────────────┐
      ▼                                 ▼
┌───────────┐                   ┌───────────┐
│ Vector    │                   │ Structured│
│ Search    │                   │ Tool (SQL/│
│ (Dense)   │                   │ CSV Engine│
└─────┬─────┘                   └─────┬─────┘
      │                               │
      └───────────────┬───────────────┘
                      ▼
       ┌────────────────────────────┐
       │ Schema & Grounding Filter  │
       └─────────────┬──────────────┘
                     ▼
       ┌────────────────────────────┐
       │ LLM Generator + Citations  │
       └────────────────────────────┘
```

5. Hardware Requirements & Deployment Targets

Production Compute Footprint

Recommended serving environment: 2x NVIDIA A10G (24GB VRAM) or 1x NVIDIA H100 GPU using vLLM for high-throughput batching, achieving up to 120 tokens/sec stream outputs.